

Internet Inequality: The Role of Gender in Digital Access in India

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1 Introduction

In the modern digital economy, internet literacy has become foundational skill for participating in education, employment and communication. The Government of India has been on the forefront to make India ‘digital ready’, right from digitizing bank accounts, to providing digital applications during the pandemic for testing and vaccination. The capacity to effectively navigate and utilize online resources has transformed from a luxury to a necessity. However, the complete usage of these platforms is essential on how digitally literate the population is. According to data from the 78th round of the National Sample Survey, just 22% of all males and 21% of all women over the age of 15 in India have some level of digital literacy, indicating that digital literacy is extremely low for both sexes. As digital literacy skills become more advanced, the gender gap widens and it appears that there is a disparity in the number of people in each sex who are proficient in using the internet across all of India (Gupta et al., 2023). These disparities are especially pronounced in developing economies, where social norms, unequal educational opportunities, and household dynamics including intra-household biases in resource allocation and parental support often favouring male members often limit women’s and girls’ ability to acquire and use digital technologies.

The present study aims to empirically examine the relationship between gender and internet use in India, focusing on whether men and women differ in their ability to use the internet for any purpose, using data from the CAMS module of the NSS survey. Essentially, we employ a binary logit model with household-level fixed effects, allowing us to isolate intra-household differences and account for unobserved heterogeneity in household environments.

2 Literature Review

With early research showing us that males were more likely to access and use digital technologies due to disparities in income, education, job roles, and interest in technology, a significant body of literature emphasizes the gendered character of the digital divide. Although these access disparities have decreased over time, gender disparities in reported self-efficacy, skill development, and usage frequency are still significant. According to Ono and Zavodny (2003), despite improvements in internet access for women in the United States by the early 2000s, they continued to use the internet less frequently and intensely than men. In low- and middle-income nations, where internet access overlaps with societal norms and household-level inequality, this gap is much more persistent. In a cross national study employing household fixed effects, Avanesian et al. (2024) showed that women were substantially less likely than men in the same family to report having even the most basic digital skills. Their study highlighted the significance of unobserved intra-household characteristics that are usually overlooked by normal regression models, such as parental beliefs, traditional gender roles, and unequal resource allocation. These conclusions are further supported by empirical studies from South Asia. According to Sultana and Imtiaz (2018) research of university students in Bangladesh, men choose gaming and amusement on the internet, while women prefer communication and e-commerce. Interestingly, they found no significant gender difference in self-assessed ICT efficacy, indicating that usage patterns may differ more because of societal expectations than technical aptitude when educational background and access are

comparable. Additionally, the study found that parental support was a significant factor in determining ICT self-confidence, supporting the notion that families, not just people, should be the focus of digital inclusion programs. When combined, these studies underscore that gendered differences in internet use cannot be understood solely through access or income but must be analyzed through a perspective that accounts for household context, social expectations, and self-perception. The application of a household fixed effects model in our study allows us to more accurately assess whether gender alone contributes to lower internet use among women, after controlling for the shared economic and cultural conditions within a household. This approach provides a more precise understanding of the digital gender divide in India.

3 Data

The dataset for this project is the Comprehensive Annual Modular Survey (CAMS) 2022-23. The survey was conducted by the National Sample Survey Office (NSSO) under the Ministry of Statistics and Programme Implementation (MoSPI), Government of India. The primary objective of CAMS was to collect data to generate indicators related to education, out-of-pocket medical expenditure, use of mobile and internet, financial inclusion, ICT skills, possession of assets, etc (NSSO, 2024). The survey employed a two stage stratified sampling design, where the first stage units were villages/blocks and the second stage units were households in both rural and urban areas. The total number of households surveyed was 3,02,086 and the total number of persons enumerated was 12,99,988. The household-level and member-level data were merged using the unique household identification to generate the required dataset for analysis (PIB, 2024).

Variables and Summary Statistics

The primary dependent variable of interest for this study is whether the individual is able to use the internet with any electronic device such as smartphone, desktop etc. for any purpose. The primary independent variable is the gender of the individual. The other independent variables are age, marital status, formal education/training status, and employment status.

Table 1: Summary of Variables used (CAMS Dataset)

Variable	Type	Categories	Summary and Categories
Internet Use	Categorical	4	Indicates the purpose of internet use. Categories: Entertainment, Information, Both, None.
Gender	Categorical	2	Records self-reported gender identity. Categories: Male, Female.
Age	Continuous	–	Reports respondent’s age in years. Used as a continuous variable.
Education (Ever Enrolled)	Binary	2	Captures whether the respondent has ever enrolled in formal education/training. Categories: Yes, No.
Employment Status	Binary	2	Reflects current employment status. Categories: Employed, Unemployed/Not in labor force.
Marital Status	Categorical	4	Indicates current marital status. Categories: Unmarried, Married, Widowed, Divorced/Separated.

In Table 1 we have a summary of the variables that are used from the CAMS dataset. Our primary variable of interest is a categorical variable with four categories but due to computational constraints,

Table 2: Variable Descriptive Statistics

Variable	Mean / Proportion
Age	30.50
Internet Literacy (Yes)	0.74
Gender (Female = 1)	0.49
Employed	0.57
Formal Education/Training	0.82
Married	0.50
Unmarried (Never Married)	0.44
Divorced/Separated	0.0042
Widowed	0.056
Observations	1,299,988

Note: Values are rounded off. Internet use is converted to a binary variable.

we change it to a binary variable. In Table 2, we have the details of variables; the mean value or the proportion. The gender-wise distribution of the Internet literacy is given Figure 1(a) ³ and gender-wise distribution of the purpose of using the internet are given by Figure 1(b) ³.

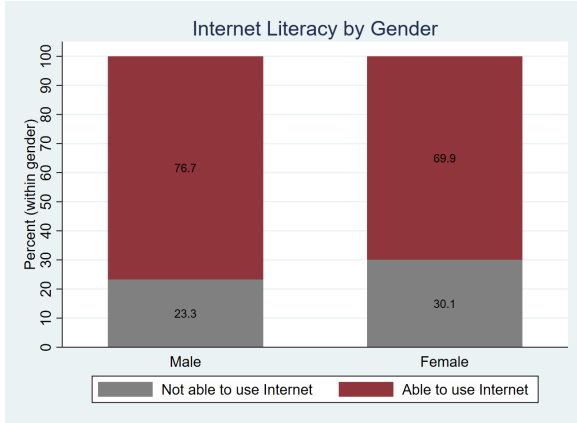


Figure 1(a): Internet Literacy by Gender

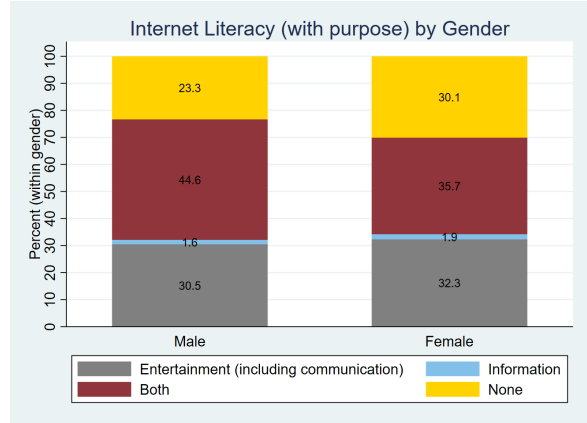


Figure 1(b): Internet Purpose by Gender

Figure 1: Internet Literacy and Purpose by Gender

4 Methodology

The baseline specification employs a logistic regression model, with internet literacy as the binary dependent variable and gender as the key independent variable. Given the binary nature of the outcome, logistic regression is an appropriate choice. The subsequent models run a conditional logistic regression, incorporating fixed effects at the household level as given in Allison (2012). Since resources are not allocated randomly within households and oftentimes take the form of gender bias against females, household-level fixed effects controls for the inter-household variation and only focuses on the intra-household variation. A fixed effects model is preferred over a random effects specification, as we assume that unobserved household-level heterogeneity, such as social norms, cultural constraints, and value systems, is likely to be correlated with the independent variables. For example, household values can influence individual decisions related to marital status or employment. The fixed effects approach also mitigates the need to control for observable household-level characteristics (e.g., religion, expenditure, or social group), as these are implicitly accounted for in the model.

Model

We estimate the likelihood of an individual being able to use the internet through a logistic regression framework. Let the binary outcome variable y take the value 1 if an individual uses the internet, and 0 otherwise. The latent variable formulation of the model is given by:

$$y^* = \mathbf{x}'\boldsymbol{\beta} + \varepsilon,$$

where y^* is the unobserved latent variable, $\mathbf{x}$ is a vector of explanatory variables, $\boldsymbol{\beta}$ is the corresponding vector of coefficients, and ε is the error term which follows a lognormal distribution with variance $\pi^2/3$. The observed binary outcome is defined as:

$$y = \begin{cases} 1 & \text{if } y^* > 0, \\ 0 & \text{if } y^* \leq 0. \end{cases}$$

The probability that an individual has the ability to use the internet, conditional on covariates $\mathbf{x}$, follows the logistic distribution:

$$\Pr(Y = 1 \mid \mathbf{x}) = \frac{\exp(\mathbf{x}'\boldsymbol{\beta})}{1 + \exp(\mathbf{x}'\boldsymbol{\beta})} = \Lambda(\mathbf{x}'\boldsymbol{\beta}),$$

where $\Lambda(\cdot)$ denotes the logistic cumulative distribution function.

The maximum likelihood estimation involves maximizing the following log-likelihood function:

$$\log \mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^N [y_i \log \Lambda(\mathbf{x}'_i \boldsymbol{\beta}) + (1 - y_i) \log (1 - \Lambda(\mathbf{x}'_i \boldsymbol{\beta}))].$$

The latent variable models are specified as follows:

$$\text{Model 1 : } y_{ih}^* = \beta_0 + \beta_1 \text{gender}_i + \varepsilon_{ih}$$

$$\text{Model 2 : } y_{ih}^* = \beta_0 + \beta_1 \text{gender}_i + \alpha_h + \varepsilon_{ih}$$

$$\text{Model 3 : } y_{ih}^* = \beta_0 + \beta_1 \text{gender}_i + \beta_2 \text{age}_i + \beta_3 \text{marital_status}_i + \beta_4 \text{education}_i + \alpha_h + \varepsilon_{ih}$$

$$\text{Model 4 : } y_{ih}^* = \beta_0 + \beta_1 \text{gender}_i + \beta_2 \text{age}_i + \beta_3 \text{marital_status}_i + \beta_4 \text{education}_i + \beta_5 \text{employment}_i + \alpha_h + \varepsilon_{ih}$$

Here, α_h denotes household-level fixed effects for the household h to which individual i belongs, and ε_{ih} is the idiosyncratic error term.

The marginal effect in a logit model is given by:

$$\frac{\partial \text{Prob}(Y = 1 \mid \mathbf{x})}{\partial \mathbf{x}} = \Lambda(\mathbf{x}'\boldsymbol{\beta}_{\text{logit}}) [1 - \Lambda(\mathbf{x}'\boldsymbol{\beta}_{\text{logit}})] \boldsymbol{\beta}_{\text{logit}},$$

The average partial effect for a dummy variable d_i is given by:

$$\frac{1}{n} \sum_{i=1}^n [\text{Prob}(Y = 1 \mid \mathbf{x}_{i(d)}, d_i = 1) - \text{Prob}(Y = 1 \mid \mathbf{x}_{i(d)}, d_i = 0)].$$

5 Results

The first model includes only gender as a predictor. The negative and statistically significant coefficient indicates that women are significantly less likely to use the internet than men. The coefficient -0.349 suggests that being female is associated with approximately 29 percent lower odds of internet literacy than males [since the odds ratio is $\exp(-0.349) = 0.705$]. The marginal effect of gender is -0.0680, which suggests that being female is associated with a 6.8 percentage point lower probability of being internet literate, compared to being male as seen in Table 4.

From the second model onwards, a conditional logistic regression model is used, that controls for unobserved household-level characteristics and gives the within-group estimates. The gender gap increases (coefficient -0.646), suggesting that within the same household, women are even less likely to use the internet than men. The marginal effect is absolutely greater (-0.1367) compared to the baseline model. This result confirms the presence of intra-household gender inequality in internet access and use.

In the third model, individual-level covariates are added. The log odds ratio (-1.40) as well as the marginal effects (-0.1560) become greater in magnitude for gender, implying women are lesser likely to

Table 3: Logistic Regression Results: Determinants of Ability to use Internet

	(1)	(2)	(3)	(4)
Female	-0.349*** (0.00459)	-0.646*** (0.00539)	-1.400*** (0.0111)	-1.835*** (0.0189)
Age			-0.128*** (0.000732)	-0.184*** (0.00125)
Married			0.947*** (0.0195)	0.380*** (0.0302)
Widowed			0.0496 (0.0468)	0.0655 (0.0741)
Divorced/Separated			0.637*** (0.0881)	-0.140 (0.124)
Has enrolled (ever) in formal education/training			1.699*** (0.0282)	1.485*** (0.0377)
Employed				0.452*** (0.0207)
Household FE	No	Yes	Yes	Yes
Observations	983128	404859	404859	346076

Note: Standard errors clustered at the household level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

be able to use internet vis-a-vis men, after controlling for individual level factors. It is also observed that factors like age have a negative effect on an individual's ability to use internet, but factors like marriage, and formal education have a positive effect as seen from Table 3 and Table 4.

In the fourth model, a dummy for employment status is added, which, by construction, reduces the sample size as individuals below the age of 15 are dropped. It is observed that for individuals of an employable age, women are less likely to be able to use the internet compared to men. However, we see that the marginal effect is reducing (-0.0680 from -0.1560 in the third model). This implies that controlling for employment status, seems to reduce the gender gap in digital literacy. Covariates like age, marriage, and education have a similar marginal effect compared to the third model. Employment status has a positive effect on digital literacy with a log odds ratio of 0.452 and a marginal effect of 0.0125.

Table 4: Marginal Effects: Determinants of Ability to Use Internet

	(1)	(2)	(3)	(4)
Female	-0.0407***	-0.1367***	-0.1560***	-0.0680***
Age			-0.0051***	-0.0128***
Married			0.0114***	0.0935***
Widowed			0.0017	0.0042
Divorced/Separated			-0.0034***	0.0598
Has enrolled (ever) in formal education/training			0.0243***	0.1140***
Employed				0.0125***
Household FE	No	Yes	Yes	Yes
Observations	983,128	404,859	404,859	346,076

Note: Marginal effects estimated after logistic regression. Each entry represents the average partial effect (dy/dx). Standard errors not shown. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Therefore, the results portray that women are significantly less likely to be internet literate than men. The gender gap widens within households and after controlling for individual factors. However, the gap narrows slightly when employment status is considered, suggesting that work participation may help reduce digital inequality.

6 Policy Implications

Based on the model results, several policy actions can be taken to reduce the gender gap in internet use. Firstly, reducing intra-household disparities requires creating digital literacy programs that actively include and favour female members, as women are much less likely than men to use the internet within the same household. Second, given that education has a significant positive impact on internet use, it is essential that girls have more access to formal education and that digital literacy be incorporated into adult and school literacy programs. Third, age-sensitive interventions, especially digital training for adult women, can assist bridge generational divides, as older people are less likely to utilize the internet. Lastly, since being employed increases the likelihood of internet use, workforce participation should be supported through skills training, job placement programs, and digital upskilling initiatives targeted at women.

7 Further Scope

This study can be enhanced to examine gender differences in the purposes for which people use the internet, going beyond just access. According to research, women prefer communication and commercial applications, but males are more likely to enjoy amusement and gaming (Sultana and Imtiaz, 2018). Using CAMS module of the NSS survey, which includes detailed indicators on the purpose of internet use, we can estimate gender-based patterns in functional usage through multinomial logit models.

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